

APPROVED



MSc Well Design and Engineering
School of Computing, Engineering and Technology

Version	2.0
Course Type	Postgraduate
Award Type	Master of Science - MSc
Awards	
No Course Award Assigned	
Code	0330
SCQF Level	Level 11
QAA Subject Benchmark	Engineering (2023)
SCQF Credit Points	180
ECTS Credit Points	90
Valid From	2026 September as of (September 2026)

HECoS

The maximum number of HECoS Codes allowed is 3.

Code	Title	Alternate Title	Percentage
100143	chemical engineering	chemical engineering	100%
			Total 100%

Professional, Statutory and Regulatory Body (PSRB)
No PSRBs have been recorded

What the Course Involves	The taught element of the course comprises four specialised modules based on Shell's Wells Distance Learning Package (WDLP) as core material. The course has the fundamental structure of four 30 credit modules (the taught element) and the 60 credit MSc Project. The MSc Project is undertaken at the end of the taught element. It is a core module, central to the objectives of the course, and enables the student to put into practice the theories taught elsewhere on the course. The MSc Stage Project is specifically intended for Shell students with considerable experience in the industry who wish to pursue a specific project relevant to their career aspirations and/or employment. The order in which the modules can be taken within each course is flexible and, it is anticipated, will depend on employment commitments. Additional Information
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	The prime client for providing sponsored delegates for this course is a major operating Company and it is expected that the delegates would attend the course from mainland Europe and other international geographical regions involved in Well Design and Engineering in the Shell Global Learning Centre in Rijswijk, The Netherlands. Access to this course is limited to students sponsored by the Shell Company or students who are ex-Shell employees but have completed Round 2 Certification.
Course Collaborations	
Shell (Award of Credit for External Provision)	
Introduction to the Course	Raising the Educational Status of Key Technical Professionals This proposal results from discussions with the Well Services discipline senior staff of Shell International Exploration and Production BV , and associate companies (SIEP) in Aberdeen and in The Hague . Shell already has a well-developed internal development programme (Round 1 and Round 2) for individuals who wish to become Well Services specialists. Within the Wells discipline key staff development is delivered using the Wells Distance Learning Package (WDLP). The proposed Course takes major elements from Round 2, and represents these as modules at Masters level. Shell Exploration and Production Europe (Shell EPE) provides in house learning modules as essential parts of the training and competence development of relevant engineering staff in the professional discipline of Well Services. The Wells Distance Learning Package (WDLP) will form a core part of the current method by which Shell EPE delivers this training to selected staff. The current Shell EPE programme offers work-based learning experience, self-study, coaching and mentoring – as well as continuous assessments (i.e. “Yellow Pages”) and examinations. The full programme links with learning administered via the Shell Open University and managed by the Wells Faculty at the Shell Global Learning Centre (SGL) in Rijswijk , The Netherlands. In addition to recognising the educational status of experienced technical staff, the Course will allow these members of staff to progress further within Shell – locally and internationally. The WDLP Round 2 is formally third party credit rated by the University, which equates to 4 x 30 SCQF11 credit rated modules (120 SCQF11 credits). This element of the course is delivered by Shell but the University is involved in quality assuring assessment and award of academic credit. The 120 credits allows entry to the MSc project component and successful completion of the 120 credits plus 60 credit project element will permit recommendation of MSc award.
	The MSc Well Design and Engineering is focused on the planning, construction, project management and delivery of wells designed to facilitate the extraction of hydrocarbons. This is a part-time course (supported by distance learning), which is aimed at those already in full time employment with Shell. In accordance with the University's statement of purpose, the aim of the Shell Corporate Programme is to enable the students to acquire the knowledge and understanding, key transferable skills, higher cognitive skills and technical skills necessary for competent practice and lifelong development as professional engineers. The course aims to develop the knowledge, understanding, capability and awareness necessary to maintain and manage existing technology at peak efficiency. It has been designed to provide a balance of technological, management and personal skills while developing leadership, innovation entrepreneurial skills in preparation for a professional career at the highest level through linkages with commerce and industry in order to prepare students for early management and leadership responsibility. The course is intended for applicants who have some employment experience in the Oil and Gas Industry, and who are seeking to progress, or redirect, their career by gaining a Masters level qualification in the Well Design and Engineering discipline that is directly relevant to their future career plans. The course is designed to give an initial broad coverage of the industry, followed by specialisation in the specific discipline.

Educational Aims of the Course	The course aims to: • Develop professional levels of competence in the range of subject areas relevant to Well Design and Engineering by building on past experience and by the reorientation of basic scientific and technological skills; • Develop skills in integrating knowledge drawn from different subject areas through coursework, case studies, industrial and project work; • Provide a deeper level of understanding of the scientific, engineering and management subjects relevant to Well Design and Engineering; • Develop a critical approach to the implementation of Well Design and Engineering; • Encourage innovative thinking, which is then applied to the discipline of Well Design and Engineering; • Produce practical Well Design and Engineering focused petroleum engineers with the high quality professional skill set required to enable, enhance and extend oil and gas field life; • Develop the ability to work within a team structure and develop a strong basic project management capability. In addition to the aims of the taught modules, the course is designed to develop initiative, creativity, interpersonal skills and a high level of critical judgment in the application of appropriate and relevant principles during the Masters level Project Stage. There is no exit award at the Postgraduate Certificate or Diploma stage for this course.
Awarding Institution	Robert Gordon University
Course Duration	Part-Time - 10 months with intakes available throughout the year.
Distinctive Features of the Course	Shell currently has in place a proficient internal development programme, the Wells Distance Learning Package (WDLP). At its core, the WDLP comprises 2 specific areas of study material, Round 1 and Round 2. Following an academic agreement between Robert Gordon University and Shell (signed May 2006), this course has been created to link the completion of Shell's Learning Programmes to university degree awards. The proposed course takes major elements from Round 2 of the WDLP, and represents these as modules credit rated at 120 SCQF level 11. This element of the course is delivered by Shell but the University is involved in quality assuring assessment. The 120 SCQF credits allows entry to the MSc project component delivered by the University and successful achievement of the 120 SCQF credits plus completion of 60 SCQF credit project element will form an MSc degree in Well Design and Engineering. Round 2 Certification Within the Shell organisation, globally, the possession of the Round 2 certificate (and the completion of the Command and Control assessment) means that the candidate has reached the competence level necessary to operate independently as a Site Supervisor. This is a critical function within the company in relation to safety, environmental management and cost management. The course has a practice-based approach, replicating the workplace environment in all of the taught modules. The course therefore allows the individual to develop a coherent Personal Learning Plan that encompasses specific modules that address particular needs, interests and aspirations within a flexible timeframe and structure. This ensures that modules can be undertaken at the relevant time in his/her career path thus maximising the potential for the subsequent learning to be integrated into the workplace therefore enhancing performance and

	future opportunities. The taught element of the course is based on Shell's Wells Distance Learning Package, offering students a balance of work based learning experience, taught and self study modules, coaching and mentoring. In addition, the course will enable the University to incorporate in-company/third party training and credit transfer. In this respect, there is the potential for modules to be developed to meet the particular training or development needs of individual organisations. The course therefore provides an effective vehicle for life long learning. This course is accredited by the Energy Institute on behalf of the Engineering Council and meets the requirements for Further Learning for registration as a Chartered Engineer. Candidates must hold a CEng accredited BEng/BSc (Hons) undergraduate first degree to comply with full CEng registration requirements."
Placement/Professional Experience	Not applicable.
Opportunities for Further Study	Further study is possible following completion of this qualification to Doctor of Philosophy (PhD) or Engineering Doctorate (EngD).
External and Internal Indicators of Quality and Standards	The QAA Engineering benchmark statement can be found at: www.qaa.ac.uk/academicinfrastructure/benchmark/statements/engineering06.asp All of the School's courses are reviewed every five years.
University Academic Regulations	This course is governed by the provisions of the University's Academic Regulations, which are available at www.rgu.ac.uk/academicregulations . In particular: Regulation A1: Courses Regulation A2: Admission and Enrolment Regulation A3: Section 1: Student Appeals (Awards and Progression) Procedure Regulation A3: Section 2: Student Conduct Procedure Regulation A3: Section 3: Student Fitness to Practise Procedure Regulation A3: Section 4: Student Fitness to Study Procedure Regulation A4: Assessment and Recommendations of Assessment Boards
University Entry Requirements	Applicants must satisfy the University's general admission requirements to courses as contained in Academic Regulation A2: Admission and Enrolment, including proficiency to a minimum standard in the English language. Specific entry requirements for this course are detailed below.
Course-Specific Entry Requirements	Admission is restricted to students entering under the Shell/RGU agreement.

Alternative Qualifications for Entry	Access to this course is limited to students sponsored by Shell and ex-Shell staff who have completed Round 2 Certification. Qualification through previous relevant industry experience may be considered. Acceptance will be subject to the University's judgment that there is a reasonable expectation that the student will be able to fulfill the learning outcomes and achieve the standard required of the award. Candidates must successfully complete the Round 2 Certificate prior to joining the MSc course.
Interview Procedures	Applicants are not usually interviewed although some may be if they do not meet the standard entry requirements.
Course-Specific Regulations	Regulation A4, paragraph 11. Compensation for failure is not permitted for the module Project for Master of Science (CEM001) which is central to the objectives of the course of study for the award of MSc. Distance learners are advised (not required) to purchase certain key textbook(s) to support their studies. It may be beneficial to also have a home computer with an Internet connection.
Equality, Diversity and Inclusion	The University has a strong commitment to the active promotion of equality across its functions, including its course provision. To this end the University has developed an inclusivity statement which articulates the University's desire to strive towards creating a working, learning and social atmosphere which is inclusive, harmonious and respectful of diversity. The University has also developed an Equality and Diversity Policy and every two years publishes a mainstreaming report, including equality outcomes, which explains how the university is developing and implementing its inclusivity statement. At a course level, Course Leaders monitor and comment on equalities data through the Annual Appraisal process. Through this process any significant trends in equalities data are highlighted. Further information can be found at: Equality & Diversity RGU. All Schools have a nominated School Contact who can advise on the requirements of the course and the range of course-specific reasonable adjustments that may be made. The University's Inclusion Centre provides support for students, details of which may be found at: The Inclusion Centre: Disability & Dyslexia Support & Advice Services RGU.
	The Purpose of the University is: • To be an innovative, inclusive and professionally focused University that has a positive impact on those we serve. The Mission of the University is: • To transform people and communities by providing excellent teaching, research and enterprise opportunities to contribute to economic, social and cultural and environmental sustainability. The Culture of the University is: • To recognise and celebrate the values of curiosity, authenticity, approachability, collaboration, enterprise, ambition, respectfulness, and equality.

University Strategy	Driven by the spirit of curiosity to continually improve, our values and principles will guide us. Our partnerships with other organisations and the communities we serve are also a mechanism to develop and grow our diversity. We have a strong commitment to widening and extending access to our courses, regardless of an individual's background. At the core of the University's significant achievements is the commitment and dedication of our people. The sense of partnership between our staff and students is strong and supporting our students to succeed in their studies and beyond is deeply ingrained in our culture. These features are reflected in the University's student satisfaction rates, which in recent years have consistently placed RGU in the top 10 of universities in the UK. RGU has a long history of inspiring highly employable graduates. This is achieved through close, collaborative contact with employers in the design and delivery of our curriculum; the range and breadth of accreditations with professional and statutory bodies; the significant role work placements and work-based learning play within our courses; and the embedding of enterprise across all our activity. In a changing society, we will ensure the continued evolution of our curriculum so that our graduates can continue to succeed in their ambitions and make a positive impact within society well into the future. To achieve our ambitions, our curriculum must be global in outlook, and we must maximise the international reach of our course portfolio. The University's research activities must have international impact and our enterprise activities must reflect the global reach of the economic sectors we work with and the ambitions of the region. RGU recognises that the creation of a sustainable future for all is a priority that unites us globally, and which demands expertise, creativity, and innovation from all subject areas and professional disciplines. Consequently, sustainability is integrated into all courses, equipping students with skills and competencies that are essential for their future careers and lives.
Course-Specific Learning Experience	The Philosophy of Course Approach to Learning This course, as a part of a suite of postgraduate courses, has been developed in close collaboration with expertise from Shell. The course has been designed to add significantly to the Continuing Professional Development (CPD) experience of students currently in employment. Students are encouraged from the outset to take control of their own learning within a framework set by the course team. The MSc Project Stage is intended to be a substantial opportunity for students to display their capability for independent learning and the acquisition of a range of professional skills appropriate to the challenges of engineering in the Oil and Gas Industry. These skills are further developed through exposure to, and use of, industry standard software tools and personal development planning. The overall strategy of the course is to provide a sound understanding of the fundamental principles and practices underpinning well design and engineering technology and to develop the specialist knowledge and interdisciplinary skills required of a professional engineer. The course involves a progressive and cumulative process of teaching, learning, assessment and feedback. In the early stages, the teaching and learning process is achieved by an appropriate combination of didactic teaching and problem solving exercises. Throughout the course emphasis is placed on student centred learning and project work. Students are encouraged to work and to present ideas in groups (as well as on an individual basis) in order to develop interpersonal skills, and a critical and evaluative approach to their work. Continued professional development and a lifelong learning ethos are emphasised as a major element of the course. The use of problem solving techniques enhances the students' understanding of the practical application of the relevant well design and engineering principles. Throughout the duration of the course, students are encouraged to take responsibility for their own learning and are supported by tutorial systems and a study support facility. A range of assessment practices is used for assessing students' learning, and is described in the Module Descriptors. Many assessments are coursework based and, given the nature of the courses, are closely keyed to case studies or to the students' work experience.

	The attachment of an academic member of staff to each module taught by Shell staff ensures consistency of academic standards of delivery and assessment.
Teaching and Learning Strategy	The course has been designed to prepare students for professional careers within the Oil and Gas Industry. As practising engineers, graduates of the course will need to be able to continue to gain and expand their knowledge and experience through the professional media available to them. The teaching and learning methods utilised within the course have been devised with this in mind. Directed and recommended reading and textbooks are used throughout, to provide basic principles and terminology. This is supplemented by the study of industry standard technical publications and papers, for instance from the Society of Petroleum Engineers (SPE). The use of such current materials ensures that the course is always up to date, cutting edge and of immediate relevance to employers. This is particularly relevant in such a dynamic and continually evolving area as the Oil and Gas Industry. The use of these materials also prepares graduates of the course to refer to and use these industry standard publications in their future careers. The Learning Process: Teaching Strategy and Methods The course curriculum is designed to enable the student to acquire and develop their subject knowledge and understanding, intellectual and thinking ability and practical and key transferable skills at level 11 corresponding to the Scottish Credit and Qualifications Framework. As the student progresses through the course they expand their understanding and critical appreciation of wells project management processes and develop skills in well design and engineering management. The overall philosophy and strategy proposed for the course is to provide a sound knowledge and understanding of well planning and design analysis within a broader context of well engineering management. The professional world relating to Well Design and Engineering technology is rapidly changing, demanding sound knowledge and advanced practical skills in order to succeed. This course will prepare innovative, informed, and resourceful graduates who are equipped for the contemporary professional environment that they enter. The course therefore allows the individual to develop a coherent Personal Learning Plan that encompasses specific modules that address particular needs, interests and aspirations within a flexible timeframe and structure. This ensures that modules can be undertaken at the relevant time in his/her career path thus maximising the potential for the subsequent learning to be integrated into the workplace and enhancing performance and future opportunities. The taught element of the course is based on Shell's Wells Distance Learning Programme, offering students a balance of work based learning experience, taught and self study modules, coaching and mentoring. Students are encouraged from the outset to take control of their own learning within a framework set by the course team (RGU and Company). The MSc Project Stage is intended to provide a substantial opportunity for students to display their capability for independent learning and the acquisition of a range of professional skills appropriate to the challenges of engineering in the Oil and Gas Industry. These skills are further developed through exposure to and use of industry-standard software tools and personal development planning. The course involves a progressive and cumulative process of teaching, learning, assessment and feedback. In the early stages, the teaching and learning process is achieved by an appropriate combination of didactic teaching, problem solving exercises and group project. Throughout the course emphasis is placed on student centred learning and project work. Students are encouraged to work and to present ideas in groups (as well as on an individual basis) in order to develop inter-personal skills, critical and evaluative approach to their work. Continued professional development and a life-long learning ethos are emphasised as a major element of the course. Throughout the duration of the course, students are encouraged to take responsibility for their own learning and are supported by tutorial systems and the study support service. Employment-Based Learning The course is aimed at those already in full time employment with Shell. The workplace is an opportunity to apply the theories of the taught modules to the professional environment and is found to be enormously beneficial to students and is particularly beneficial in developing ideas and theories for the MSc Project Stage.

	The Student's Role in Learning The course will be offered via the Part-Time mode, supported by Distance Learning (DL). The main mode of directed study is by network distance learning through Shell Global Learning Centre (SGL). This allows the exchange of information between the students providing them with the ability to ask questions relating to the study. Discussion threads enable students to share experience, reinforce learning, and contextualise material. In addition, students will undertake several web-based activities as an integrated part of their learning process. As online learners, students are part of a “virtual cohort” and will be considered independent learners. As a result, they will be required to take the responsibility of where and how learning occurs and to communicate with fellow members of their “cohort”. All students are expected to be active participants in the learning process. This means that there is a requirement for students to extend their learning from formally delivered materials through wider reading, research self-directed enquiry. The student is expected to utilise the full range of resources available to them, in addition to information searching techniques taught on the course in support of all their studies. As he or she develops over the duration of the course, the student is expected to become increasingly motivated and self-supporting, taking greater control of their studies and defining his or her personal learning. The student is encouraged to regularly reflect on their learning and to establish a view of their own progress and development which may form the basis of tutorial guidance with staff. The main vehicle for independent learning is the personal research based MSc Project.
Assessment	The Assessment of Student Work All courses employ a range of strategies to achieve the intended learning outcomes. Consonant with the Masters level it is the students' responsibility to elicit the most appropriate academic and practical help for personal progress. This means that teaching and learning strategies are responsive to individual needs, yet also encourage group interaction. Various forms of assessment are used to monitor student progress. Module assessment requirements are specified in the corresponding Module Descriptor. It has been recognised that for Masters level courses it is not always possible to assess students' critical and analytical skills at the appropriate level through traditional time limited, unseen examinations. As a result of this, all the modules are assessed by a combination of coursework and exam. Each assessment involves summative feedback, in the form of a written response which is given by tutors to the student for future development and reference. Taught Modules The taught element of the course is based on the major elements of Wells Distance Learning Package (WDLP) as core material. The WDLP is divided into volumes, sections, parts and topics with a set of questions at the end of each part or topic. These questions (historically referred to as the “Yellow Pages”) are an integral part of the distance learning process. Successful completion of the “Yellow Pages” will also provide evidence of awareness, knowledge and skills of the students' wells competence portfolio. In all modules students are required to submit course work to provide evidence of achievement of the learning outcomes, as indicated in the Module Descriptors. In addition, this evidence will be assessed by way of an examination on completion of each module. Assessment of the MSc Project

	The assessment is a portfolio submission including a Thesis and presentation of findings. Students will learn to critically evaluate primary and secondary data, critically analyse project outcomes, consider ethics and sustainability and identify and communicate strengths and limitations. Where appropriate, the project will include a substantial practical application of technical skills.
University Support for Teaching, Learning and Assessment	The University provides a number of means of supporting teaching and learning: • student induction organised on a course or School basis; • a Team Around a Student (TAAS) system of pastoral support to students including online support www.rgu.ac.uk/policies [The TAAS School Student Support Model replaces the Personal Tutor model of student support from Session 2026-27]; • an extensive programme of student study skills delivered through Student Life and the Library; • an extensive library of learning resources; • close collaboration with industry and professional, statutory and regulatory bodies; • the University's active participation in Scotland's Tertiary Enhancement Programme (STEP) www.step.ac.uk ; • support for staff from the Department of Academic Quality and Learning Enhancement (AQLE), including credit-rated teaching, learning and assessment provision; • extensive opportunities for student placements with companies or organisations, academic overseas exchange programmes, and support for entrepreneurial activity and community engagement opportunities; • access to a range of professionally qualified, quality assured careers professionals to support individual students and graduates career planning and opportunity search, supported by a range of bespoke and contemporary online advice and information including the extensive RGU eHub; • Moodle, http://campusmoodle.rgu.ac.uk , the University's dedicated virtual learning environment; • a commitment to knowledge exchange and technology transfer through focused research activity, which contributes to the critical underpinning for all taught courses; • the expanding provision of state-of-the-art, purpose-built facilities and buildings.
	Delivery Mode As explained previously, the taught modules form part of the WDLP. This package has been developed by the Shell Global Learning Centre in the Netherlands and is distributed to candidates world-wide. Each candidate is assigned to a local, specialist mentor (where applicable). Support is also available on-line from Holland and so the course is, effectively, delivered by a combination of part-time and by distance learning. For the purposes of this documentation, the part-time mode is used for simplicity as it fits more closely the mode of delivery. All modules within the programme are offered in the Distance Learning (DL) mode. Taught modules will be delivered from the Shell Global Learning Centre (situated in the Netherlands) to Shell operation Units worldwide. It forms a significant part of the teaching and learning process and is used for posting of lecture notes and activity questions. Although the University's Virtual Learning Environment (Campusmoodle) will not be used to deliver course material, it will be utilised as an additional support tool by the students.

Course-Specific Support for Teaching, Learning and Assessment

Wells Global Network

Distance Learning courses are delivered from the Shell Global Learning Centre (SGL). An area has been set up within the Wells Global Network (WGN) specifically for those students participating on the Wells Distance Programme. This allows the exchange of information between the students providing them with the ability to ask questions relating to the study of the Wells Distance Learning Package. Discussion threads enable students to share experience, reinforce learning, and contextualise material.

Shell Open University provides a generic infrastructure for the management and support of online learning. Facilities to be provided include:

- course registration system;
- discussion forums;
- message servicing allowing student feedback and support from supervisors/mentors;
- calendar of course events;
- tracking of student interaction with resources provided via the Virtual Campus;
- provision of web based open learning materials and other appropriate resources.

It also contains detailed information on the learning process. It contains information and links to other areas such as the WGN and Exploration and Production (EP) learning website.

Teaching of taught modules will be facilitated by staff from Shell Global in collaboration with additional colleagues from third party Companies contributing to specific specialised topics in modules (for example input on new technologies).

The learning experience centres on the creation and sustenance of active community groups, and delivery of topics is structured in such a way as to facilitate the maintenance of student interaction. The Shell Global library provides support for distance learners and is progressively expanding its provision of electronic texts such as e-journals and books.

Resources and Student Support

Students will be supported through a variety of ways, including (but not limited to):

- Induction programmes;
- Student Handbooks and Module Guides;
- Library and Study Skills Programmes and Package;
- Virtual Learning Environment;
- Immediate supervisors within Shell;
- Mentors assigned by Shell;
- Colleagues.

In addition to the Virtual Learning Environment (Campusmoodle) there are a number of sources of support for students.

The Shell Global Learning Centre (SGL) has a wide range of onsite facilities including laboratories, model rooms and drilling and production simulators to facilitate student learning.

Shell also operates a web-based course management system, TeleTOP. This allows students to add contributions to the overall course environment and includes web links and suggestions for further reading and resources. The WDLP assessment sections (historically referred to as the "Yellow Pages") can be submitted via this system which in turn allows the students' mentors to respond with appropriate feedback.

Further support will come from the staff of the Shell Support Centre within the School.

The MSc Project

Project Definition and Management

The main vehicle for independent learning is the personal research based Project. The student will receive the Project Handbook and training/instruction in the following basic aspects of methodology: the nature of research; writing a research proposal; sourcing relevant information; data collection; analysis and synthesis of data; research ethics and writing up their work.

Masters students are expected to develop their own project proposals. This generates "real" company project work and gives the students the motivation to complete the research successfully. In many cases the student will have their own ideas for a potential project - perhaps drawn from current work experience. At an early stage, the student should establish detailed objectives for the work involved.

The MSc Project work is based on the following principles:

- the Project is rated at 60 Credits and is equivalent to 600 hours full-time work;
- the student is expected to identify a project and a Company Supervisor;
- a formal project specification is drawn up;
- the Project Specification and supervisory arrangements are approved by the School of Engineering and an Academic Supervisor appointed;
- the Project is assessed by a portfolio submission including a Thesis.

Masters projects fall naturally into subject areas related to oil and gas industry e.g. Well Design and Engineering. Within these broad areas are specific projects. A masters student is expected to identify an industrial project and a Supervisor. An RGU Academic Supervisor from the School is then attached to the project. The Academic Supervisor has the responsibility for keeping a watchful eye on the project development, for giving guidance to the student and Company Supervisor where appropriate and in acting as a Co-examiner in the assessment process.

The student project is defined in a Project Specification (as described in the MSc Project Handbook). This states the aims, objectives and constraints associated with the project to all concerned. The responsibilities of the student, Company Supervisor and the Academic Supervisor are also clearly defined in the MSc Project Handbook.

The Project Specification document is developed by the student in conjunction with the Company Supervisor and submitted to the Programme Leader in the School. It then goes through the process of "School moderation" to ensure that:

- the Project Specification is relevant to the course aims and objectives;
- the project aims are feasible;
- the subject has sufficient potential and an appropriate challenge (in terms of level and volume) for a Masters degree;
- suitable supervisory arrangements are in place;
- the resources required to support the project are available within Shell, or accessible elsewhere;
- any confidentiality issues and ethics issues have been considered.

The School has a library of past project reports, which students are welcome to look at and also borrow if they wish. The University's Virtual Learning Environment (Campusmoodle) will be utilised as an additional support tool for the students.

University Monitoring of Quality and Standards	The University employs several mechanisms for evaluating and improving the quality and standards of teaching, learning and assessment, including: • <i>Annual Course Appraisal Reports</i> are prepared for each course and reviewed and approved by Course/Programme Management Teams and School Academic Boards which consider, amongst other things, feedback generated from student questionnaires; • <i>Institution-Led Subject Review</i> , involving external panel members, on a six-yearly basis to formally review its major subject provision, followed by a three year interim review to monitor progress against actions/issues raised through the review process; • <i>External Examiner Annual Reports</i> ; • Ongoing liaison with industrial/professional liaison groups. Formal Committees with responsibility for monitoring and evaluating quality and standards: • Staff/Student Liaison Committees, or equivalent; • Course/Programme Management Teams; • School Academic Boards; • Assessment Boards; • Quality Assurance and Enhancement Committee and associated Sub-Committees.
Course-Specific Monitoring of Quality and Standards	As noted in the Course Aims, the WDLP is closely linked to the relevant competency matrix. The taught element, therefore, is closely related to industry requirements.
University Student Feedback Mechanisms	The University and RGU:Union work in partnership to create an environment which stimulates: • the participation of students, by empowering them to proactively provide views and opinions; • a supportive learning community where students and staff engage in meaningful dialogue; and • the engagement of students in the design of solutions and enhancements. This approach is underpinned by the <i>Student Partnership Agreement</i> which promotes the ethos of partnership at all levels of the University. Further information can be found at the Student Representation and Partnership Moodle Area: Student Representation & Partnership RGU . There are a variety of opportunities for students to provide feedback to the University, and to become actively involved in shaping their learning experience. These opportunities are integral to the University's approach to the quality assurance and enhancement of teaching and learning, and the holistic student experience. Mechanisms through which students are engaged and supported include: • participation in student evaluation questionnaires and where relevant the <i>National Student Survey (NSS)</i> ;

	• staff/student liaison arrangements at course/programme level; • support for Student Representatives; • student representation on Institution-Led Subject Review Panels and Validations; • student representation on key University committees.
Course-Specific Student Feedback Mechanisms	Staff/Student Liaison The School operates a Staff Student Liaison Committee for its Postgraduate Programmes. The committee meets at least once per semester. The Postgraduate Programme Management Team receives comments and suggestions from the Student Liaison Committee as a standing agenda item. The course will be delivered through the Shell Global Learning Centre, the Netherlands. As a result of this, students will be given the opportunity to discuss issues relating to all aspects of the learning experience with course staff via a discussion forum. Staff will be required to respond to all issues raised with this information being circulated to all students within the course in a timely manner. Course/Module evaluation feedback will be gathered for review appraisal and development purposes through the SGL. The Shell Learning Centre has a quality system already in place to allow for student feedback.

Awards

Stage 1

<i>Permissible Awards</i>	<i>Title</i>	<i>Location</i>	<i>Target Award</i>	<i>PSRB</i>	<i>Award Credits</i>
Master of Science - MSc	Well Design and Engineering	Aberdeen, United Kingdom	Target Award		180

Level Learning Outcomes

On successful completion of this Course the learner will be able to :

#	Description	SCQF Characteristic	SCQF Level
LLO1	To apply advanced mathematics, natural science and engineering principles to analyse and solve well design and engineering problems. [M1]	SC1 SC2	SCQF 11, SCQF 11
LLO2	To formulate, analyse and evaluate well design and engineering problems using appropriate scientific reasoning, engineering judgement, and interpretation of incomplete or uncertain data. [M2]	SC1 SC2 SC3	SCQF 11, SCQF 11, SCQF 11
LLO3	To select and apply suitable computational and analytical techniques to model well design and engineering scenarios and critically evaluate the limitations of these methods. [M3]	SC1 SC2 SC3 SC4	SCQF 11, SCQF 11, SCQF 11, SCQF 11
LLO4	To select and critically evaluate technical literature, industry standards, and other information sources to solve and support decision-making in well design and engineering problems. [M4]	SC1 SC2 SC3	SCQF 11, SCQF 11, SCQF 11
LLO5	To design original, practical and responsible engineering solutions that meet technical, operational, societal, customer and commercial needs within well design and engineering. [M5]	SC1 SC2 SC3 SC5	SCQF 11, SCQF 11, SCQF 11, SCQF 11
LLO6	To evaluate the environmental, societal and lifecycle impacts of oil and gas production and propose solutions that minimise adverse effects. [M7]	SC1 SC2 SC5	SCQF 11, SCQF 11, SCQF 11
LLO7	To work effectively as individuals and as members or leaders of teams, evaluating their own performance and contributing constructively to collaborative well design and engineering projects. [M16]	SC2 SC5 SC4	SCQF 11, SCQF 11, SCQF 11
LLO8	To communicate well design and engineering concepts clearly and effectively to technical and non technical audiences through a range of professional methods and formats. [M17]	SC4	SCQF 11

Programme Outcome Types Reference		
#	SCQF Characteristic	SCQF Level
SC1	Knowledge and understanding	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC2	Practice: Applied knowledge, skills and understanding	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC3	Generic cognitive skills	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12

SC4	Communication, numeracy and IT skills	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12
SC5	Autonomy, accountability and working with others	SCQF 7 - SCQF 7 SCQF 8 - SCQF 8 SCQF 9 - SCQF 9 SCQF 10 - SCQF 10 SCQF 11 - SCQF 11 SCQF 12 - SCQF 12

Delivery Range

<i>Title</i>	<i>Description</i>
Part Time Online	Route A - Post-2005 Students. Route B - Pre-2005 Students.

Stage 1 / Semester 1**Part Time Online**

Route A - Post-2005 Students. Route B - Pre-2005 Students.

Core								
<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
Part Time Online	ENM252	Casing Design		Paula Rorie (abspmcrr)	Yes	1.0	30	Level 11
Part Time Online	ENM251	Drilling Well Control		Paula Rorie (abspmcrr)	Yes	1.0	30	Level 11
Part Time Online	CEM001	Project for Master of Science		Ibiye Iyalla (ii5282)	Yes	1.0	60	Level 11
Part Time Online	ENM250	Well Engineering		Paula Rorie (abspmcrr)	Yes	1.0	30	Level 11
Part Time Online	ENM253	Wells Project Management		Paula Rorie (abspmcrr)	Yes	1.0	30	Level 11

Course Deliveries

<i>Stage</i>	<i>Semester</i>	<i>Delivery Type</i>	<i>Delivery Range</i>	<i>Code</i>	<i>Title</i>	<i>Notes</i>	<i>Owner</i>	<i>Allow Mapping</i>	<i>Version</i>	<i>Credits</i>	<i>Module Level</i>
1	Semester 1	Core	Part Time Online	ENM252	Casing Design		Paula Rorie (abspmcrr)	Yes	1.0	30	Level 11
1	Semester 1	Core	Part Time Online	ENM251	Drilling Well Control		Paula Rorie (abspmcrr)	Yes	1.0	30	Level 11
1	Semester 1	Core	Part Time Online	CEM001	Project for Master of Science		Ibiye Iyalla (ii5282)	Yes	1.0	60	Level 11

1	Semester 1	Core	Part Time Online	ENM250	Well Engineering		Paula Rorie (abspmcrr)	Yes	1.0	30	Level 11
1	Semester 1	Core	Part Time Online	ENM253	Wells Project Management		Paula Rorie (abspmcrr)	Yes	1.0	30	Level 11

Delivery

Location	Mode of Study	Mode of Attendance	Teaching Period Key Date
All Applicable Locations	Online	Part Time	

Sources of Further Information

No sources of further information recorded
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